

⚠ IMPORTANT: PROPOSED BACKEND ARCHITECTURE & SCHEMA FREEZE NOTICE

This document presents the **proposed backend architecture, database schema, and content management fields** for the new Orkay website. Please review all fields, section capabilities, and workflows carefully. **If you require any additions, deletions, or structural modifications, please let us know what to change right now during this review stage.** Once this specification is approved and backend provisioning (DynamoDB keys, GSIs, and API contracts) commences, the data architecture will be locked and **post-development structural modifications cannot be made** without a separate Change Request.

CLIENT ENTITY	HOSTING & DATABASE	CATALOG CAPACITY
Orkay Tiles Pvt. Ltd. (Morbi, Gujarat)	AWS (Amplify + DynamoDB + S3, Mumbai)	4,000+ to 10,000+ SKUs

1. EXECUTIVE SUMMARY

This document defines the functional requirements, field schemas, media pipelines, and role-based permissions for the **Custom AWS Backend CMS** for Orkay Tiles. The system provides zero-subscription, Orkay-owned infrastructure designed for rapid catalog operations, paperless B2B dealer onboarding with Aadhaar e-Sign, and high-performance digital asset delivery.

2. USER ROLES & PERMISSION MATRIX (2-TIER RBAC)

FEATURE / ADMIN MODULE	SUPER ADMIN (MANAGEMENT / IT)	GENERAL EDITOR (CATALOG & MARKETING)
Product Catalog: Add, Edit, Delete, Categorize & Publish	Full Access	Full Access
Bulk Catalog Import: Excel / CSV Upload & Validation	Full Access	Full Access
Marketing Content: Hero Banners, Milestones, Spaces, Stats	Full Access	Full Access
Download Center: PDF Catalogs, Spec Sheets & Certifications	Full Access	Full Access
Dealer Application Viewer & Lead Export: CSV / GHl sync	Full Access	View Only
Dealer KYC Approval & Digio Aadhaar e-Sign Issuance	Full Access	No Access
Staff User Management, Password Resets & Audit Logs	Full Access	No Access

3. PRODUCT CATALOG CMS FIELDS (4,000+ SKUS)

Structured schema for live search, faceted multi-attribute filtering, and rich technical display.

FIELD NAME	TYPE	REQUIRED	OPTIONS / FORMAT	EXAMPLE VALUE
Product Name	Text (String)	Mandatory	Max 100 chars	Statuario Bianco
SKU / Code	Alphanumeric	Mandatory	Unique Master Key	OKY-STA-1218
Category	Dropdown	Mandatory	Vitrified Porcelain / Slab / Ceramic / Outdoor	Vitrified Porcelain Tiles
Available Sizes	Multi-Select	Mandatory	600×600, 600×1200, 1000×1000, 1200×1800, 800×2400, 800×3000, 300×450, 300×600 mm	600×1200, 1200×1800
Surface Finish	Multi-Select	Mandatory	Glossy, High Gloss, Matt, Silk Matt, Carving, Rustic Sugar, Bookmatched, Anti-Skid	Glossy, Bookmatched
Look / Pattern	Dropdown	Mandatory	Marble Look, Statuario, Stone Look, Travertine, Concrete, Wood, Exotic Onyx, Terrazzo	Statuario Look
Surface Application	Multi-Select	Mandatory	Floor, Wall, Facade, Counter Top, Heavy Commercial, Wet Areas / Bath	Floor, Wall, Counter Top
Colour Family	Multi-Select	Mandatory	White, Beige, Grey/Charcoal, Black, Brown, Blue, Green, Gold/Warm	White / Off-White, Grey
Tile Body Type	Dropdown	Mandatory	Glazed Vitrified (GVT/PGVT), Full Body, Color Body, Ceramic	Glazed Vitrified (PGVT)
Faces / Randoms	Dropdown	Optional	1 Face, 2 Faces, 4 Faces, 6 Faces, 8+ Faces, Continuous Bookmatch	4 Faces

Field Name	Type	Required	Options / Format	Example Value
Thickness (mm)	Number	Mandatory	Standard thickness (e.g. 9.0, 9.5, 10.5, 12.0, 15.0 mm)	9.5 mm
Water Absorption	Number	Mandatory	ISO / EN compliance (e.g. < 0.05%, < 0.5%)	< 0.05%
Packaging Details	Group Fields	Mandatory	Pieces / Box, Sq. m / Box, Sq. ft / Box, Weight (kg)	2 pcs · 2.88 sq m · 58 kg
Description	Rich Text	Optional	Application highlights & sales copy (Max 500 words)	Detailed editorial copy
Primary Tile Image	Upload (1:1)	Mandatory	1:1 Aspect ratio, Min 3000×3000 px, Web-ready	face_oky_sta_1218.webp
Gallery Images	Multi-Upload	Optional	Room / Ambient / Close-up shots (Max 6)	room_oky_1218_01.webp
Spec Sheet PDF	Doc Upload	Optional	Technical specification & test report PDF (Max 5 MB)	spec_oky_sta_1218.pdf
Status & Badges	Toggles	Mandatory	Active/Inactive, Featured on Home, New Launch, Bestseller	Active New Launch

4. SECTION-BY-SECTION EDITABLE WEBSITE CONTENT

Marketing staff can independently update key sections of the website without touching code.

Website Section	Editable Fields	Media Specifications	Live Behavior
Hero Section	- Main Headline & Sub-headline - Primary CTA Text & Target Link - Media Selector (Video vs. Still Image)	Video: 1600×900 VP9/WebM (<7 MB) Still: 2560×1440 WebP (<400 KB)	Instant homepage refresh with video fallback
Corporate Statistics Bar	4 Metric Cards: - Numerical Value (e.g. 60,000) - Suffix Text (e.g. sq m produced daily) - Subtitle Label	None (Text & Numerical values)	Live dynamic counter animation
Campus & Infrastructure	For each unit (Showroom, 7 Units, Design Studio, Dispatch Yard, Mundra Port): Title, Subtitle, Feature Image	Image: 1200×800 WebP (3:2 Aspect Ratio)	Horizontal swipe carousel
Architectural Spaces	For each space (Lobby, Living, Kitchen, Bath): - Room Title, Description Copy - Primary Hero Photo & Sub-gallery	Image: 1600×1000 WebP (16:10 Ratio)	Interactive lookbook slider
Milestone Timeline	For each year (1996 to 2025+): - Milestone Year & Capacity Claim - Machinery / Technology & Formats - Archival Photo	Image: 800×600 WebP (4:3 Ratio)	Chronological history scroll
Download Center	For each Catalog / Brochure: - Title, Category Tag, Cover Thumbnail - PDF Document Upload	Thumbnail: 600×800 WebP PDF: Max 25 MB	Direct download link with auto file-size badge
Company Certifications	For each Certificate (ISO, CE, IEC): - Title, Issuing Authority, Expiry Date - Badge Icon & Full Certificate PDF	Badge: Transparent PNG/WebP PDF: High-res vector PDF	Footer & trust-bar display

5. TWO-STAGE DEALER ONBOARDING & AADHAAR E-SIGN

A streamlined digital acquisition funnel converting prospective dealers into verified, legally contracted partners.

STAGE 1: Public Express Inquiry (Website Form)

Public Lead

Captures applicant eligibility: Firm Name, Contact Person, Mobile Number (+91 OTP verification), State & District, Current Showroom Carpet Area (sq ft), and Annual Turnover estimate.

ORKAY INTERNAL REVIEW (CMS Dashboard)

Admin Gate

Orkay sales management verifies district vacancy (*One District, One Dealer* policy). With one click, admin triggers secure Stage 2 KYC invitation link via SMS and WhatsApp.

STAGE 2: Secure KYC Portal & Document Upload

Private Portal

Dealer uploads GSTIN Certificate, Company PAN, Showroom & Godown Photos (3–5 clear images), Bank Details & Cancelled Cheque, and Authorized Signatory Aadhaar info.

DIGIO AADHAAR E-SIGN & AUTOMATED DEALER CODE ISSUANCE

Legally Binding

Signatory completes instant Aadhaar-OTP e-Signature under IT Act 2000. System automatically reserves and issues unique Dealer Code (e.g. `ORK-DLR-GJ-042`) and logs tamper-proof audit trail in DynamoDB.

5.1 Dealership Administration Fields (Internal CMS View)

ADMIN FIELD NAME	FIELD TYPE	OPERATIONAL PURPOSE
Application Status	Dropdown	Applied → Shortlisted → KYC Submitted → e-Sign Sent → Active → Rejected
Assigned Dealer Code	Alphanumeric (Unique)	System-generated unique identifier (e.g. <code>ORK-DLR-GJ-042</code>)
Allotted Territory / District	Dropdown	Guarantees territory exclusivity under the <i>One District, One Dealer</i> model
Approved Commercial Terms	Text / Currency	Credit limit & payment terms (e.g. <code>₹15,00,000 – 30 Days PDC</code>)
Security Cheque Reference	Text (String)	Bank Cheque Number or Security Fixed Deposit receipt number
Assigned Area Sales Manager	Dropdown	Maps the internal Orkay sales manager responsible for servicing the dealer

6. DIGITAL ASSET LIFECYCLE & REPLACEMENT RULES

Core Principle: Speed, Redundancy, and Zero Broken Links

Images uploaded to the CMS are automatically optimized for mobile speeds while protecting against accidental deletions.

- What happens when an image is replaced?** Every file is assigned a unique content-hash filename (`prod_a9f8e7d6_1200.webp`). When an image is replaced, the system generates a new hash URL, instantly bypassing old CDN caches and guaranteeing visitors immediately see the latest image without delay.
- 30-Day Safe Recovery (Trash Archive):** Old or replaced images are never permanently deleted on day one. They are automatically moved to an S3 trash partition (`s3://orkay-media/archive-trash/`) with an automated **30-Day AWS Lifecycle Auto-Purge Rule**. If a file is replaced by mistake, it can be fully restored within 30 days.
- Automated Multi-Resolution Derivative Generation:** Staff upload only one high-resolution master file. A background AWS Lambda function automatically generates 4 WebP derivatives:

DERIVATIVE TYPE	RESOLUTION	FORMAT	WEBSITE PLACEMENT	TARGET WEIGHT
Grid Card	600 × 600 px	WebP	Catalog product grid & filter listings	40–60 KB
Detail Main	1200 × 1200 px	WebP	Product detail main display viewport	120–250 KB
Zoom Lightbox	2400 × 2400 px	WebP	Fullscreen tile texture inspection	400–600 KB
Hero Full-Bleed	2560 × 1440 px	WebP	Desktop widescreen hero banners	300–500 KB
Archive Master	3840 px / Original	JPEG q90	Glacier Instant Retrieval (Archive only)	2–5 MB

7. ADDITIONAL SYSTEM DATA STORED IN THE CMS

- Website Inquiries & WhatsApp Leads:** Full record of customer leads, inquiry timestamps, and real-time sync status with GoHighLevel (GHL) CRM.
- Immutable Audit Trail:** Logs every staff action (who created/modified SKU, who approved dealer, who updated banner, with exact timestamps).
- 301 SEO Redirect Manager:** Maps legacy WordPress URLs to the new Next.js routes to preserve all historic Google search rankings.

8. CLIENT ACTION ITEMS & REVIEW CHECKLIST

- ☐ **Q-01 (Product Categories):** Confirm the 4 product categories (*Vitrified Porcelain Tiles, Porcelain Slab Tiles, Ceramic Tiles, Outdoor Tiles*) for launch.
- ☐ **Q-02 (Dealer Onboarding Documents):** Confirm if any additional compliance documents (e.g. Trade Reference, Shop Act) are required for your dealer network.

- ☐ **Q-03 (Digio Agreement Draft):** Supply the approved standard Dealership Agreement template (Word/PDF) with signature placeholder tags for the Digio e-Sign integration.
- ☐ **Q-04 (Admin Accounts):** Nominate the email addresses of the initial Super Admin and General Editors who will receive CMS access credentials.
- ☐ **Q-05 (Excel SKU Dataset):** Prepare the initial product dataset using the provided import template with at least 3 high-resolution web-ready images per design.